Ben-Gurion University of the Negev

Faculty of Engineering Science

School of Electrical and Computer Engineering

Dept. of Electrical and Computer Engineering

Fourth Year Engineering Project

PDR
Project name – AskUni: A Smart Assistant for Student Life

Project number:	**p-2026-037**	
Students (name & ID):		
	Doron Wizman	**207579608**
	Gil Friedman	**211862149**
Supervisors:	**Dan Vilenchik**	
Sponsors:		
Submitting date:	**14/11/2025**	

Table of Contents

1. Abstract 1
    1.1 Hebrew Abstract 1
2. The purpose of this project 2
3. Development and Work Phases 2
    3.1 Text Collection and Requirements Analysis 2
    3.2 Data Preprocessing and Embedding Generation 2
    3.3 Vector Database and Semantic Search 2
    3.4 LLM Integration and Answer Generation 2
    3.5 User Interface and System Testing 2
    3.6 Evaluation and Expansion 2
4. Approaches to Solving the Problem 3-4
    4.1 Manual Centralization of Websites 3
    4.2 Keyword-Based Search Engine 3
    4.3 AI-Driven Natural Language Interface (Selected Approach) 3
    4.4 Off-the-Shelf Commercial Solutions 3-4
5. Proposed Solution 4
6. Technology and innovations to solve the problem 4
7. Flowchart illustration 4
8. literature review 5
9. Bibliography 5

1 Abstract

In recent years, the university's digital ecosystem has become increasingly fragmented, encompassing multiple platforms such as faculty websites, course registration systems, personal portals, and administrative interfaces. This fragmentation has created an inconsistently structured information environment, where effective navigation often requires prior institutional knowledge and considerable time investment. The challenge is particularly pronounced for new users - such as incoming students and prospective applicants - who frequently struggle to locate essential forms, procedures, and academic resources. Although the university website currently offers a basic keyword-based search function, its performance and accuracy remain limited, often yielding incomplete or poorly structured results.

This shortcoming underscores the need for a more intelligent and context - aware navigation framework. To address these challenges, this project proposes the development of an AI - driven intelligent navigation and search system that enables seamless interaction with the university's digital resources. Leveraging LLMs and Vector Search technologies, the system will analyze and index web content across university platforms, interpret user queries in natural language, and provide precise, contextually relevant responses. The goal is to enhance user experience by reducing search time, improving information accessibility, and enabling more efficient exploration of the university's online environment.
Keywords: LLM, Vector Search, Smart Navigation.

1.1 תקציר

בשנים האחרונות הפכה סביבת המידע הדיגיטלית של האוניברסיטה למבוזרת ומורכבת יותר, וכוללת מגוון רחב של פלטפורמות כגון אתרי פקולטות, מערכות רישום לקורסים, פורטלים אישיים וממשקים מנהליים. פיצול זה יצר סביבה שבה מבנה המידע אינו עקבי, והניווט בין המערכות דורש ידע מוקדם על מבנה האוניברסיטה והשקעת זמן משמעותית. הקושי בולט במיוחד בקרב משתמשים חדשים- כגון סטודנטים חדשים ומועמדים ללימודים אשר מתקשים באיתור טפסים, נהלים ומשאבים אקדמיים חיוניים.

למרות שהאתר האוניברסיטאי כולל כיום מנוע חיפוש בסיסי המבוסס על מילות מפתח, ביצועיו מוגבלים והוא מספק לעיתים תוצאות חלקיות או לא ממוקדות. מגבלה זו מדגישה את הצורך במסגרת ניווט חכמה ומבוססת הקשר. כדי להתמודד עם אתגרים אלו, מוצע הפרויקט הזה לפתח מערכת חכמה לניווט וחיפוש מבוססת בינה מלאכותית ,שתאפשר אינטראקציה טבעית עם מקורות המידע הדיגיטליים של האוניברסיטה.

באמצעות שילוב בין (LLM) ו- Vector Search המערכת תבצע ניתוח ואינדוקס של תכנים מאתרי האוניברסיטה, תפרש שאילתות בשפה טבעית ותספק תשובות מדויקות ורלוונטיות להקשר.

מטרת המערכת היא לשפר את חוויית המשתמש, לצמצם את זמני החיפוש, להגדיל את הנגישות למידע, ולאפשר ניווט יעיל ואינטואיטיבי יותר במערכות המקוונות של האוניברסיטה.

מילות מפתח: LLM, חיפוש וקטורי, ניווט חכם.

2 The purpose of this project

This tool can be applied to several use cases. Its primary application is intelligent navigation across the university's fragmented web platforms, providing a seamless and unified way to access information distributed among different university systems.

The project aims to create a unified, AI-driven interface that enables users to interact with the university's online systems through natural language queries.

By integrating Large Language Models (LLMs) with Vector Search technology, the system will be able to understand user intent and provide relevant, accurate, and concise answers drawn from multiple sources within the university's digital ecosystem.

Ultimately, the project seeks to enhance the overall user experience, reduce search time, and improve accessibility and efficiency when navigating university resources.

3 Development, and Work Phases

This section outlines the main phases of the system's development, from data collection and processing to deployment and evaluation.

3.1 Text Collection and Requirements Analysis

Collect relevant content from official university sources and analyze user needs to define main system requirements.

3.2 Data Preprocessing and Embedding Generation

Clean and segment text data into semantic units, remove irrelevant content, and generate embeddings representing each segment's meaning.

3.3 Vector Database Construction and Semantic Search

Store embeddings in a vector database (e.g., FAISS, Pinecone) and implement semantic search for meaning-based retrieval with optimized response time.

3.4 LLM Integration and Automated Answer Generation

Integrate an LLM (e.g., GPT) with the vector search to produce coherent, context-aware responses and deploy a functional AI chatbot on the university website.

3.5 User Interface Development and System Testing

Develop an intuitive web interface, conduct user testing, and refine the system for improved accuracy and usability.

3.6 Results Evaluation and Feature Expansion

System performance will be evaluated based on answer accuracy, response time, and overall efficiency compared to traditional keyword-based search. Following evaluation, advanced navigation features such as direct links and step-by-step guidance will be added, and system coverage will be expanded to additional departments and user groups.

4 Approaches to Solving the Problem

4.1 Manual Centralization of Websites

A straightforward solution would involve manually integrating all existing university portals into a single, centralized website.

Advantages: Simple architectural concept and No need for complex AI algorithms.

Disadvantages: Requires extensive coordination among departments and High maintenance costs and limited scalability.

4.2 Keyword-Based Search Engine

Another possible approach is developing an internal search engine that indexes all university websites and retrieves pages based on keyword matches. This approach is currently being implemented at Ben Gurion, but it does not solve the problem well, and therefore this approach was once accepted and is currently undergoing adaptation.

Advantages: Easy to implement using existing web indexing tools and Provides quick answers to simple and infrequent queries (when there aren't many results for a word).

Disadvantages: Does not understand context or intent behind natural-language questions , Returns many irrelevant results and Produces irrelevant or incomplete results when terminology differs from official phrasing.

4.3 AI-Driven Natural Language Interface (Selected Approach)

The chosen approach utilizes Large Language Models (LLMs) combined with Vector Search to create an intelligent, semantic search system. The system converts all website content into embeddings that represent the meaning of each text segment.

When a user submits a query in natural language, it is also converted into an embedding and compared to stored vectors to find the most semantically relevant content .

Finally, an LLM formulates a clear, concise, and context-aware response.

Advantages: Understands natural language and user intent , Provides accurate, context-based results even without keyword matches and Scalable and adaptable to additional university systems.

Disadvantages: Requires computing resources for embedding and model inference , Needs periodic retraining and data synchronization and Requires more work than a Keyword-Based Search Engine.

4.4 Off-the-Shelf Commercial Solutions

Several commercial tools currently offer AI-based search and chatbot functionalities that could theoretically assist in navigating large digital ecosystems.

Advantages: These commercial solutions are mature, reliable, and provide ready-to-use integration for large organizations with lots of money.

Disadvantages: They are costly, require vendor-managed infrastructure, and are not adaptable to academic environments such as universities. Specifically, they cannot easily integrate with systems like course registration, Moodle, or faculty websites at Ben-Gurion University.

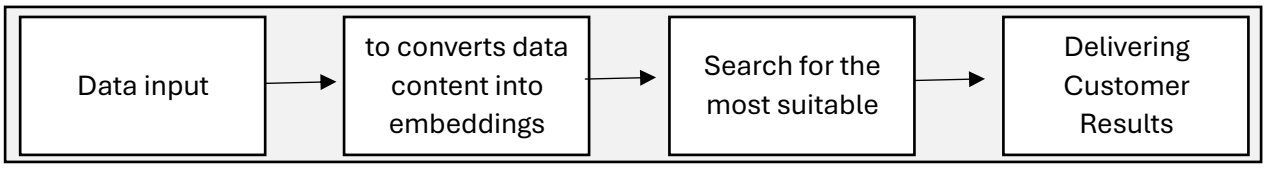
5 Proposed Solution

The proposed solution is to design and develop an AI-based intelligent navigation assistant that enables students to access university information using natural language queries.
The system will use a Large Language Model (LLM) combined with Vector Search technology to understand user queries, process university web content, and deliver context-aware and accurate answers in real time.
The system will operate as follows:
The system will collect and process data from university web sources, convert it into semantic embeddings for efficient search, interpret natural language queries using an LLM to identify intent, retrieve the most relevant information through vector similarity search, and generate concise, accurate answers via a user-friendly chatbot or smart search interface.

6 Technology and innovations to solve the problem

The project is built on advanced AI technologies, including Large Language Models (LLMs) for natural language understanding and Vector Search for semantic information retrieval. Additionally, it leverages modern Python libraries for data collection, processing, and integration across the university's decentralized systems.
Together, these technologies enable intelligent, efficient, and context-aware navigation within the academic digital ecosystem.
Project Innovation: The core innovation is not just the use of AI, but its specific application:
Intent-Based Search: Transforming the current keyword-based search into a semantic, conversational system that understands user intent.
Platform Unification: Creating a single, intelligent interface that operates on top of the university's fragmented digital systems, saving users time.
A Custom & Cost-Effective Alternative: Unlike expensive, generic commercial products, this project introduces an open, cost-effective solution specifically optimized for BGU's academic content and workflows.

7 Flowchart illustration



8 literature review

Prior work on academic information access centers on domain-specific chatbots and FAQ assistants. A BotPress/NLP enrolment bot achieved ~90.6% accuracy, showing guided dialogue helps routine queries but relies on predefined Q&A/keywords and a single-site scope [1]. For scholarly portals, a journal-website chatbot answered 3,000 queries in six months-reducing staff workload but exposing accuracy drift, governance issues, and the need for staged training/human oversight [2]. Web-crawler dialogue-tree agents with Seq2Seq + BiLSTM + attention improve intent/entity extraction (82.1% vs 63–76% baselines; 87.1% in-domain, beating RasaNLU) yet still require curated per-site trees and lack cross-platform semantic retrieval [3]. With LLMs and vector databases, retrieval now goes beyond keywords: a recent survey details embedding pipelines that enable RAG and natural-language navigation across fragmented ecosystems [4]. In academia, LLM-driven vector search unifies access to course, program, and faculty information, letting users ask in natural language and receive context-aware answers [5].

Building on this, our project combines LLM-based query understanding with vector search over multi-platform university content to (i) overcome keyword/Q&A brittleness, (ii) unify disparate systems without manual dialogue-tree construction, and (iii) add governance—source attribution, data privacy, feedback loops—to curb hallucinations and sustain accuracy. Yet existing studies remain single-domain or KB-bound; none address cross-platform semantic retrieval integrated with LLM-based reasoning. This project bridges that gap with a unified, AI-driven navigation system for complex university environments.

9 Bibliography

[1] Meera, S. N., Naga Malleswari, K. S., Sandeep, N., Srikanth, V., & Bindu, B. H. (2025). Smart FAQ Chatbot with Voice Assistant Driven by AI. International Journal of Communication Networks and Information Security, 17(3), 127–130.

[2] Abdulla, A. I., Mohammed, I. J., Yousif, Z. Y., & Alsubari, B. (2024). Generative AI Chatbot for Engineering Scientific Journal. Tikrit Journal of Engineering Sciences, 31(3), 72–79. https://doi.org/10.25130/tjes.31.3.7

[3] Hsih, M.-H., Yang, J.-X., & Hsieh, C.-C. (2024). Design and Implementation of an Intelligent Web Service Agent Based on Seq2Seq and Website Crawler. Information, 15(12), 818. https://doi.org/10.3390/info15120818

[4] Wang, J., Asai, A., Pan, L., Han, X., et al. (2024). When Large Language Models Meet Vector Databases: A Survey. arXiv:2402.01763.

[5] Khalife, G., Durugbo, C., & Smith, S. (2024). An LLM Approach to Course Recommendations Using Natural Language Queries. arXiv:2412.19312.

הערכה לשיחת סקר תכנון ראשוני ((PDR

<u>אם יש צורך, לכל סטודנט/ית בנפרד</u>

מספר הפרויקט: 20-P_____- שם הפרויקט:

שם המנחה החיצוני:

שם המנחה מבית הספר:

שם הסטודנט/ית: ת.ז:.

%	קריטריון	1-חלש	2-בינוני	3-מצוין
20	הגדרת המטרה - האם מטרת הפרויקט ברורה? האם הוגדרו קריטריונים ברורים ומדידים כמותית לבחינת הצלחת הפרויקט			
20	הבנת הבעיה - האם ניכר שהסטודנטים מבינים את הבעיה ואת הפתרונות הקיימים? האם נערך סקר ספרות רלוונטי?			
20	הצעת הפתרון - האם הגישה הכללית ורכיבי הפתרון הוצגו והוסברו בצורה ברורה? האם הובנו מהות העשייה, הצורך וסביבת היישום?			
20	בדיקת היתכנות - האם הסטודנטים ביצעו מחקר בסיסי לבדיקת היתכנות פתרון לבעיה? כמו כתיבת קוד, ניתוח מעגל, ניתוח מודל מתמטי..			
20	איכות כתיבת ה PDR - תחביר, הגדרת המטרה, סקר ספרות, תקצירים, מפרט טכני)או הצעת מחקר(והצעת תכנון מפורטים.			

הערות: